

Daily Geometry Drill #6

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: equal angles standing on the same chord · $PA \cdot PB$ products · Ptolemy $AC \cdot BD = AB \cdot CD + BC \cdot AD$ · Brahmagupta $K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ · Pitot for tangential quadrilaterals.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Spot the chord, then either double the angle at the centre or equalise the angle in the same segment.

- In a circle with centre O , the angle at the circumference subtending arc AB is 20° . Write down the angle at the centre.
- Points P, Q, R, S lie on a circle, and $\angle PQR = 35^\circ$. Write down $\angle PSR$.
- $ABCD$ is a cyclic quadrilateral with $\angle A = 70^\circ$ and $\angle B = 85^\circ$. Write down $\angle C$.
- The tangent at T to a circle makes an angle of 40° with the chord TA . By the alternate segment theorem, write down the angle in the alternate segment subtended by TA .
- Chords AB and CD of a circle intersect inside at P , with $PA = 2$, $PB = 6$ and $PC = 3$. Write down PD .
- A tangent from P touches the circle at T , and a secant through P cuts the circle at A then B . Given $PT = 3$ and $PA = 1$, write down PB .
- AB is a diameter of a circle and P is any other point on the circle. Write down $\angle APB$.
- Two secants from the external point P cut the circle as $P-A-B$ and $P-C-D$, with $PA = 4$, $PB = 6$ and $PC = 2$. Write down PD .
- A tangent from P has length 8, and a secant through P meets the circle at A (nearer) and B , with $PA = 4$. Write down PB .
- In the cyclic quadrilateral $ABCD$, $\angle BAC = 30^\circ$. Write down $\angle BDC$.

Section B Manipulation Drills

(≤50 s each)

Half the central angle, subtract cyclic opposites from 180° , or multiply chord pieces – then pick from the options.

1. O is the centre of a circle and $\angle AOB = 130^\circ$. The angle at the circumference subtending AB is?
 - A) 60°
 - B) 65°
 - C) 70°
 - D) 130°
2. $ABCD$ is cyclic with $\angle A = 100^\circ$. Then $\angle C$ is?
 - A) 80°
 - B) 100°
 - C) 90°
 - D) 160°
3. Chords AB and CD cross at an interior point P , with $PA = 3$, $PB = 5$ and $PC = 15$. Then PD is?
 - A) 1
 - B) 3
 - C) 5
 - D) 9
4. A tangent from P has length 5, and a secant through P gives outer segment $PA = 2$. Write down PB (the full secant segment).
5. A tangent at T makes an angle of 55° with chord TA . The angle in the alternate segment subtended by TA is?
 - A) 35°
 - B) 55°
 - C) 110°
 - D) 70°
6. The secant PAB from an external point has $PA = 2$, $PB = 18$. The length of the tangent PT drawn from P is?
 - A) 6
 - B) 9
 - C) 12
 - D) 36
7. $ABCD$ is cyclic with $\angle A = 2x$ and $\angle C = 3x$. Find x .
 - A) 24

- B) 30
C) 36
D) 72
8. AB is a diameter of a circle and C is a point on the circle with $\angle ABC = 55^\circ$. Write down $\angle BAC$.
9. A *tangential* (circumscribed) quadrilateral $ABCD$ has $AB = 5$, $BC = 7$ and $CD = 6$. By Pitot's theorem ($AB + CD = BC + DA$), find DA .
A) 4
B) 6
C) 8
D) 18
10. Chords AB and CD intersect at P with $PA = 6$, $PB = 4$ and $PC = 3$. Write down PD .

Section C Substitution & Structure(≤ 75 s each)

Chain two circle ideas per question: a tangent and a chord, two secants, or Ptolemy's theorem with its products.

1. OA and OB are radii with $\angle AOB = 100^\circ$, and a tangent touches the circle at A . The angle between this tangent and the chord AB is?
A) 40°
B) 50°
C) 60°
D) 80°
2. From the external point P , the secants PAB ($PA = 3$, $PB = 12$) and PCD ($PC = 4$) cut the circle. The full segment PD is?
A) 9
B) 10
C) 12
D) 36
3. $ABCD$ is cyclic with diagonal AC splitting $\angle BAD$ into $\angle BAC = 30^\circ$ and $\angle CAD = 40^\circ$. Then $\angle BDC$ is?
A) 30°
B) 40°
C) 70°
D) 110°

4. In a cyclic quadrilateral, $AB \cdot CD = BC \cdot AD = 15$ and the two diagonals sum to 11. What is the larger diagonal?
- A) 5
B) 6
C) 30
D) 11
5. A cyclic quadrilateral has sides 5, 5, 6, 6. By Brahmagupta's area formula for cyclic quadrilaterals, its area is?
- A) 25
B) 30
C) 36
D) 60
6. AB is a diameter of a circle and D is another point on the circle. Write down $\angle ADB$.
- A) 60°
B) 90°
C) 45°
D) 120°
7. From the external point P , the tangent PT has length 6 and the secant PAB has nearer point A with $PA = 3$. What is the chord length AB ?
- A) 6
B) 9
C) 12
D) 18
8. $ABCD$ is cyclic with $\angle A : \angle C = 2 : 3$ and $\angle B : \angle D = 5 : 4$. Find $\angle A$.
- A) 60°
B) 72°
C) 100°
D) 108°

Section D Challenge Ramp(≤ 90 s each)

Stack power of a point, Ptolemy and Brahmagupta into coordinated diagrams – each answer is a single clean number.

1. Chords AB and CD cross interiorly at P with $PA \cdot PB = PC \cdot PD = 30$, and PD is one more than PC . Find PC .
- A) 4

- B) 5
C) 6
D) 30
2. In a cyclic quadrilateral, $AB \cdot CD = 18$ and $BC \cdot AD = 24$. One diagonal is 7. What is the other diagonal?
A) 4
B) 5
C) 6
D) 42
3. A cyclic quadrilateral has sides 2, 3, 4, 1. Its area (Brahmagupta) is?
A) $2\sqrt{6}$
B) $\sqrt{24}$
C) 6
D) $2\sqrt{3}$
4. $ABCD$ is cyclic; the diagonal AC splits $\angle BAD$ into 30° and 40° . What is the angle $\angle BCD$ opposite $\angle BAD$?
A) 70°
B) 110°
C) 140°
D) 90°
5. The isosceles trapezoid with legs 5 and bases 6 and 12 is cyclic (every isosceles trapezoid is cyclic). Its area, via Brahmagupta (semiperimeter 14), is?
A) 36
B) 24
C) 40
D) 18

“A circle is an equality machine: every theorem just asserts two angles or two products are the same, and your eye should find the chord holding them together.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths